Multiple Testing and Clustering

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Outline

Where We Left Off

- Yesterday we ran DESeq2 on the `airway` dataset and got a results table with one row per gene.
- Two columns matter most:
  - `pvalue`: the raw p-value from the Wald test for that gene.
  - `padj`: the adjusted p-value printed in the output.
- Today we answer: why are these different, and which one should you use?
- The short answer: `padj`. Always.
- The longer answer requires understanding the multiple testing problem.

The Multiple Testing Problem

- In a standard hypothesis test at $\alpha = 0.05$, there is a 5% chance of a false positive when H_0 is true.
- That is fine for one test. What about 20,000?
- With M independent tests all under H_0:

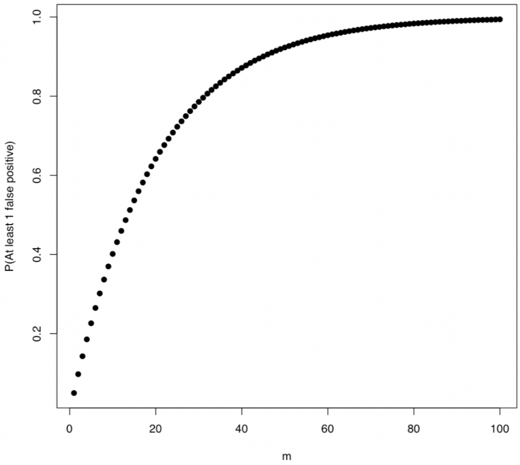
$$P(\text{at least one false positive}) = 1 - (1 - \alpha)^M$$

- At $\alpha = 0.05$ and $M = 20{,}000$:

$$1 - (0.95)^{20000} \approx 1$$

- Virtually certain to produce false positives.
- A more concrete way to think about it: if all 20,000 genes are non-differentially expressed and we use raw p-values at $\alpha = 0.05$, we expect $0.05 \times 20{,}000 = 1{,}000$ false positives.

Probability of At Least 1 False Positive



Two Ways to Control the Error Rate

- Two main error rates to control:
- **Family-wise error rate (FWER):**

$$\text{FWER} = P(\text{at least one false positive})$$

  Appropriate when any false positive is costly. Conservative.
  - Example: a clinical trial testing three primary endpoints. One false positive could lead to approving an ineffective treatment.

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- **False discovery rate (FDR):**

$$\text{FDR} = E\left(\frac{\text{false positives}}{\text{all rejections}}\right)$$

  Appropriate when testing many hypotheses and some false positives are acceptable, as long as most discoveries are real.
  - Example: an RNA-seq study identifying candidate genes for follow-up. A few false leads are tolerable; missing many real signals is not.

Bonferroni: The Conservative Option

- The simplest FWER procedure: divide α by the number of tests.
- Reject H_j if $p_j \leq \alpha/M$.
- Equivalently, multiply each p-value by M (cap at 1):

$$\tilde{p}_j = \min(Mp_j,\ 1)$$

- **Example:** $M = 20{,}000$, $\alpha = 0.05$. A gene must have a raw p-value below $0.05/20{,}000 = 2.5 \times 10^{-6}$ to be called significant.
- **When to use it:** small number of planned comparisons where any false positive has serious consequences.
- **When not to use it:** genomics. The threshold is so stringent that almost nothing passes, and many true signals are missed. This is not caution; it is a different kind of error.

The Benjamini-Hochberg (BH) Procedure

- Controls the FDR at level α (DESeq2 uses 0.05 by default).
- Procedure:
  1. Sort raw p-values: $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(M)}$.
  2. Find the largest rank k such that $p_{(k)} \leq \frac{k}{M}\alpha$.
  3. Reject all hypotheses with rank $1, 2, \ldots, k$.
- If we declare all genes with `padj` < 0.05 significant, we expect at most 5% of those calls to be false positives.
- This is far less conservative than Bonferroni.
- In R : `p.adjust(pvalues, method = "BH")` gives adjusted p-values. DESeq2 computes them automatically.

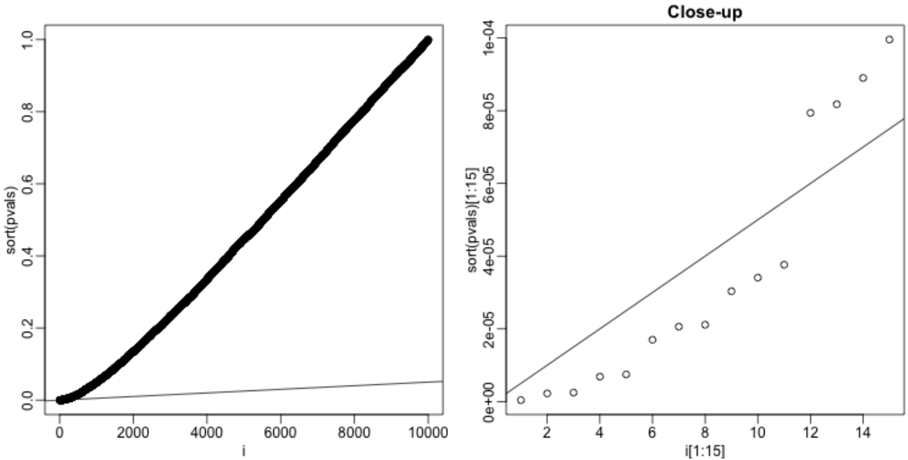
BH Procedure: Worked Example

$M = 10$ tests, $\alpha = 0.05$.

Rank j	$p_{(j)}$	$\frac{j}{M}\alpha$	$p_{(j)} \leq \frac{j}{M}\alpha$?	Reject?
1	0.0008	0.005	Yes	Yes
2	0.009	0.010	Yes	Yes
3	0.016	0.015	No	Yes*
4	0.019	0.020	Yes	Yes
5	0.196	0.025	No	No
6	0.350	0.030	No	No
7	0.541	0.035	No	No
8	0.781	0.040	No	No
9	0.900	0.045	No	No
10	0.993	0.050	No	No

*We reject all hypotheses up to and including rank $k = 4$ (the highest rank where the condition is met), even though rank 3 did not individually satisfy it.

BH Procedure: Geometric Picture



FWER vs FDR: Which to Use?

	FWER (Bonferroni)	**FDR (BH)**
Controls	$P(\geq 1$ false positive$)$	Expected % false discoveries
Setting	Few planned tests	Many simultaneous tests
Conservative?	Very	Moderately
Power	Low	Higher
Genomics?	Rarely appropriate	Standard practice
Clinical trial?	Often appropriate	Sometimes

Practical guidance:

- In RNA-seq and genomics: use FDR control (BH). DESeq2 does this automatically. Report genes with `padj` < 0.05.
- In a small study with a handful of pre-specified comparisons: Bonferroni or no correction may be defensible.
- When in doubt: report both raw and adjusted p-values and let the reader judge.

Multiple Testing in DESeq2

- DESeq2 applies the BH procedure automatically when you call `results()`.
- The `padj` column is the BH-adjusted p-value.
- Two additional things DESeq2 does:
  1. **Independent filtering:** genes with very low mean expression are excluded from testing entirely. Testing them would require extreme evidence to reject H_0 and would only inflate the multiple testing burden. Removing them increases power for the genes that remain.
  2. **Outlier detection:** genes with a single sample that has an extreme count (Cook's distance outlier) get $\texttt{pvalue} = \texttt{padj} = \texttt{NA}$.
- Both procedures are conservative: they can only increase power for the genes that remain, never create false positives.

Why Clustering?

- Multiple testing tells us which genes are differentially expressed. But 500 significant genes is hard to interpret one at a time.
- Do the significant genes fall into groups with similar expression profiles? Do samples cluster by treatment group?
- **Clustering** is an unsupervised method: we are not trying to predict a known outcome. We are looking for hidden structure.
- In genomics, clustering is used for:
  - Grouping genes with similar expression patterns across samples (rows of the count matrix).
  - Grouping samples with similar overall expression profiles (columns).
  - Identifying patient subtypes in clinical studies.

K-Means Clustering

- **Goal:** partition n observations into K clusters so that observations within a cluster are as similar as possible.
- **Algorithm:**
  1. Choose K. Randomly assign each observation to one of the K clusters.
  2. Compute the centroid (mean) of each cluster.
  3. Reassign each observation to the cluster with the nearest centroid.
  4. Repeat steps 2–3 until assignments stop changing.
- The algorithm minimizes the total within-cluster sum of squares:

$$\text{minimize} \sum_{k=1}^{K} \sum_{i \in C_k} \|x_i - \bar{x}_k\|^2$$

- In `R` : `kmeans(x, centers = K, nstart = 25)`

K-Means: Things to Know

- **You must choose K.** This is the central difficulty.
- **Scale your data first.** K-means uses Euclidean distance, so variables with large variance dominate. Use `scale()` before clustering.
- **Results depend on initialization.** Different random starts can give different final clusters. Always use `nstart > 1`.
- **K-means assumes spherical clusters of roughly equal size.** It works poorly when clusters are elongated, non-convex, or very different in size.
- **Sensitive to outliers.** A single extreme observation can pull a centroid toward it and distort the clustering.
- In genomics, K-means is often applied to the top principal components (from the Day 11 PCA) rather than the raw count matrix.

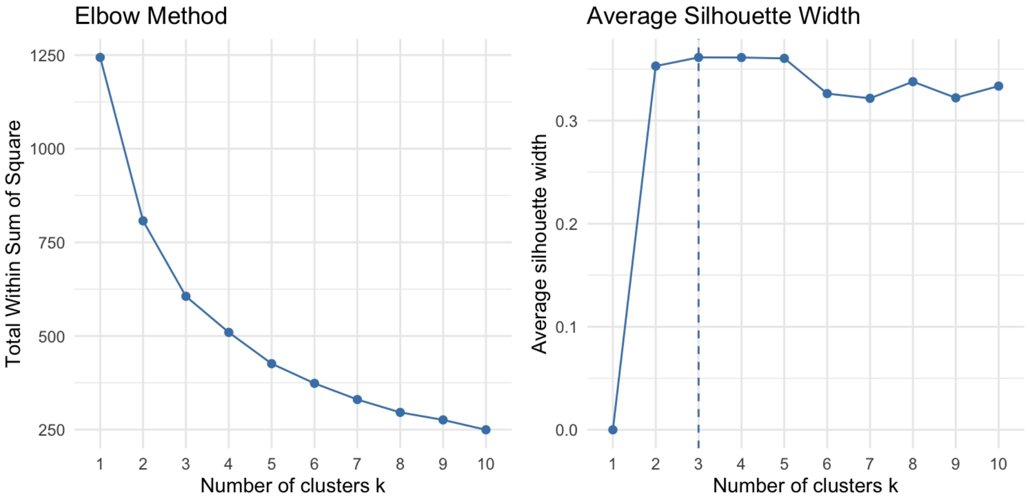
Choosing K: The Elbow Method and Silhouette Width

- **Elbow method:** plot total within-cluster sum of squares (WSS) against K. Look for a bend (elbow) where adding more clusters gives diminishing returns.
- **Silhouette width:** for each observation i:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i),\, b(i)\}}$$

  where $a(i) =$ avg distance within cluster, $b(i) =$ avg distance to nearest other cluster.
- $s(i)$ ranges from -1 to $+1$. Higher is better.
- Average silhouette width S summarizes the whole clustering. Choose K that maximizes S.

Choosing K: The Elbow Method and Silhouette Width

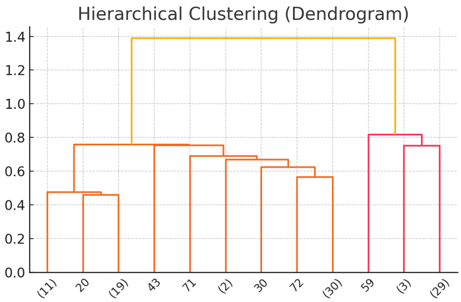


Hierarchical Clustering

- K-means requires choosing K upfront. Hierarchical clustering does not. It builds a tree (dendrogram) showing all possible groupings from n clusters down to 1.
- **Agglomerative** (bottom-up) algorithm:
  1. Start: each observation is its own cluster.
  2. Find the two most similar clusters and merge them.
  3. Repeat until all observations are in one cluster.
- The result is a tree. Cut it at any height to get any number of clusters.
- Three common **linkage methods** that define distance between clusters:
  - **Complete:** maximum distance between any two points in the clusters. Tends to produce compact, roughly equal-sized clusters. Default choice.
  - **Average:** average distance between all pairs.
  - **Single:** minimum distance. Tends to produce long chain-like clusters (chaining effect).

The Dendrogram

- Each leaf is one observation.
- The height at which two branches merge indicates how dissimilar those groups are. Earlier merges (lower height) = more similar.
- To get K clusters: draw a horizontal line across the dendrogram and count the branches it crosses.



The Dendrogram

- **In genomics:** the heatmap from Day 12 used hierarchical clustering to order both the rows (genes) and columns (samples). The dendrograms along the edges of the heatmap are exactly this.
- In `R` :
  - `dist(x)` computes the distance matrix.
  - `hclust(d, method = "complete")` fits the tree.
  - `cutree(hc, k = K)` cuts to K clusters.

K-Means vs. Hierarchical Clustering

	K-Means	**Hierarchical**
Must specify K upfront	Yes	No
Produces a dendrogram	No	Yes
Scales to large n	Well	Poorly ($O(n^2)$ memory)
Sensitive to outliers	Yes	Less so
Assumes cluster shape	Spherical	None
Result deterministic	No (random init)	Yes
Use in genomics	Top genes / PCs	Sample clustering, heatmaps

In practice:

- Use hierarchical clustering for samples (few columns in the count matrix) to check whether treatment groups separate.
- Use K-means (or PCA + K-means) for genes (many rows) when looking for co-expression patterns.

Clustering Is Exploratory

- Clustering is an exploratory tool, not a formal inferential one.
- There is no p-value for a cluster. There is no test of whether the clusters you found are "real."
- Every dataset will produce some cluster structure – even random data. The question is whether the structure is scientifically meaningful.
- Ways to build confidence in a clustering result:
  - Does the same structure appear with different methods (e.g., K-means and hierarchical)?
  - Does the structure replicate in an independent dataset?
  - Does the cluster membership correlate with known biological or clinical variables that were not used to build the clustering?
- **The heatmap is your friend.** Visualizing the clustering alongside the raw data lets you see whether the groups actually look different, not just whether an algorithm says they are.

Review: Key Points

Multiple testing:

- Testing M hypotheses inflates false positives. Adjustment is required.
- FWER (Bonferroni): controls probability of any false positive. Very conservative. Rarely appropriate in genomics.
- FDR (BH procedure): controls expected proportion of false discoveries. Standard in RNA-seq. Use `padj`.

Clustering:

- K-means: fast, requires K upfront, sensitive to scaling and outliers. Use `nstart > 1`. Choose K with elbow or silhouette.
- Hierarchical: produces a dendrogram, no K needed upfront, used to order heatmap rows and columns. Complete linkage is the default.
- Clustering is exploratory. Validate findings with biological knowledge, replication, and visualization.

Tomorrow: PCA and penalized regression in the genomics context.